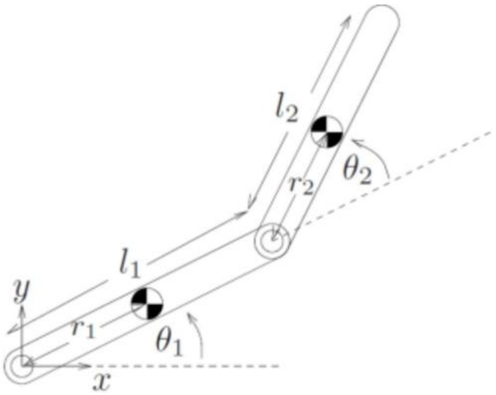
Homework 6

(Due time:8:30,Apr.30,2026)

1. For the two-link planar manipulator shown below.



Model each link as a homogenous rectangular bar with mass $m_1 = m_2 = 12$Kg, length $l_1 = l_2 = 1.0$m, $r_1 = r_2 = 0.5$m and moment of inertia tensor

$$I_1 = I_2 = \begin{bmatrix} 0.0125 & 0 & 0 \\ 0 & 1.0025 & 0 \\ 0 & 0 & 1.01 \end{bmatrix}$$

relative to a frame attached at the center of mass of the link and aligned with the principle axes of the bar.

(a) Derive the dynamic equations of the manipulator (using Lagrange's Equations)

(b) Derive the dynamic equations of the manipulator (using computation algorithm of Dynamics of Open-chain Manipulators with the Christoffel Symbol)

(a) 平面机械臂，Z轴转动惯量 $I_{z_1} = I_{z_2} = 1.01\ kg\cdot m^2$

连杆1质心：

$x_1 = r_1\cos\theta_1 \qquad y_1 = r_1\sin\theta_1$

$\dot{x}_1 = -r_1\sin\theta_1\cdot\dot{\theta}_1 \qquad \dot{y}_1 = r_1\cos\theta_1\cdot\dot{\theta}_1$

$v_1^2 = \dot{x}_1^2 + \dot{y}_1^2 = r_1^2\dot{\theta}^2$

连杆2质心：

$x_2 = l_1\cos\theta_1 + r_2\cos(\theta_1+\theta_2)$

$y_2 = l_1\sin\theta_1 + r_2\sin(\theta_1+\theta_2)$

$\dot{x}_2 = -l_1\sin\theta_1\cdot\dot{\theta}_1 - r_2\sin(\theta_1+\theta_2)(\dot{\theta}_1+\dot{\theta}_2)$

$\dot{y}_2 = l_1\cos\theta_1\cdot\dot{\theta}_1 + r_2\cos(\theta_1+\theta_2)(\dot{\theta}_1+\dot{\theta}_2)$

$$v_2^2 = \dot{x}_2^2 + \dot{y}_2^2 = l_1^2\dot\theta_1^2 + r_2^2(\dot\theta_1+\dot\theta_2)^2 + 2l_1r_2\cos\theta_2\,\dot\theta_1(\dot\theta_1+\dot\theta_2)$$

系统总动能＝平移动能＋绕质心的旋转动能

$$T = \left(\frac{1}{2}m_1v_1^2 + \frac{1}{2}I_{z1}\dot\theta_1^2\right) + \left(\frac{1}{2}m_2v_2^2 + \frac{1}{2}I_{z2}(\dot\theta_1+\dot\theta_2)^2\right)$$

代入v_1^2, v_2^2，展开得

$$T = \frac{1}{2}(m_1r_1^2 + I_{z1})\dot\theta_1^2 + \frac{1}{2}m_2\left[l_1^2\dot\theta_1^2 + r_2^2(\dot\theta_1+\dot\theta_2)^2 + 2l_1r_2\cos\theta_2(\dot\theta_1^2 + \dot\theta_1\dot\theta_2)\right]$$
$$+ \frac{1}{2}I_{z2}(\dot\theta_1+\dot\theta_2)$$

$$V = m_1gy_1 + m_2gy_2 = m_1gr_1\sin\theta_1 + m_2g(l_1\sin\theta_1 + r_2\sin(\theta_1+\theta_2))$$

拉格朗日函数 $L = T - V$

拉格朗日方程 $\tau_i = \frac{d}{dt}\left(\frac{dL}{d\dot\theta_i}\right) - \frac{dL}{\partial\theta_i}$

$$\frac{\partial L}{\partial\dot\theta_2} = (m_2r_2^2 + I_{z2})(\dot\theta_1+\dot\theta_2) + m_2l_1r_2\cos\theta_2\,\dot\theta_1$$

$$\frac{d}{dt}\left(\frac{dL}{d\dot\theta_2}\right) = (m_2r_2^2 + I_{z2})(\ddot\theta_1+\ddot\theta_2) + m_2l_1r_2\cos\theta_2\,\ddot\theta_1 - m_2l_1r_2\sin\theta_2\,\dot\theta_1\dot\theta_2$$

$$\frac{\partial L}{\partial\theta_2} = \frac{\partial T}{\partial\theta_2} - \frac{\partial V}{\partial\theta_2} = -m_2l_1r_2\sin\theta_2(\dot\theta_1^2 + \dot\theta_1\dot\theta_2)$$

$$\tau_2 = (m_2r_2^2 + I_{z2} + m_2l_1r_2\cos\theta_2)\ddot\theta_1 + m_2r_2^2 + I_{z2})\ddot\theta_2$$
$$+ m_2l_1r_2\sin\theta_2\,\dot\theta_1^2 + m_2gr_2\cos(\theta_1+\theta_2)$$

$$\frac{\partial L}{\partial\dot\theta_1} = (m_1r_1^2 + I_{z1} + m_2l_1^2 + m_2r_2^2 + I_{z2} + 2m_2l_1r_2\cos\theta_2)\dot\theta_1$$
$$+ m_2r_2^2 + I_{z2} + m_2l_1r_2\cos\theta_2]\dot\theta_2$$

$$\frac{\partial L}{\partial\theta_1} = -m_1gr_1\cos\theta_1 - m_2gl_1\cos\theta_1 - m_2gr_2\cos(\theta_1+\theta_2)$$

$$\tau_1 = (m_1r_1^2 + I_{z1} + m_2l_1^2 + m_2r_2^2 + I_{z2} + 2m_2l_1r_2\cos\theta_2)\ddot\theta_1$$
$$+ (m_2r_2^2 + I_{z2} + m_2l_1r_2\cos\theta_2)\ddot\theta_2 - m_2l_1r_2\sin\theta_2(2\dot\theta_1\dot\theta_2 + \dot\theta_2^2)$$
$$+ (m_1r_1 + m_2l_1)g\cos\theta_1 + m_2gr_2\cos(\theta_1+\theta_2)$$

代入 $m_1, m_2, l_1, l_2, r_1, r_2, I_{z1}, I_{z2}$ 计算可得

$$\tau_1 = (20.02 + 12\cos\theta_2)\ddot\theta_1 + (4.01 + 6\cos\theta_2)\ddot\theta_2 - 6\sin\theta_2(2\dot\theta_1\dot\theta_2 + \dot\theta_2^2)$$
$$+ 18g\cos\theta_1 + 6g\cos(\theta_1+\theta_2)$$

$$\tau_2 = (4.01 + 6\cos\theta_2)\ddot\theta_1 + 4.01\ddot\theta_2 + 6\sin\theta\,\dot\theta_1^2 + 6g\cos(\theta_1+\theta_2)$$

(b) 克里斯托费符号与标准矩阵法

$$M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + N(\theta) = \tau$$

定义3个常数：

自身惯量与质量之和 $\alpha = m_1 r_1^2 + I_{z1} + m_2 l_1^2 + m_2 r_2^2 + I_{z2} = 20.02$

耦合惯量 $\beta = m_2 l_1 r_2 = 6.0$

连杆2的本身惯量 $\delta = m_2 r_2^2 + I_{z2} = 4.01$

① 提取广义惯性矩阵 $M(\theta)$

$$T = \frac{1}{2}\dot{\theta}^T M(\theta)\, \dot{\theta}$$

根据(a)中T的表达式提取出 $\dot{\theta}_1^2, \dot{\theta}_2^2, \dot{\theta}_1\dot{\theta}_2$ 的系数

可以写出 $M(\theta)$

$$M(\theta) = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} = \begin{bmatrix} \alpha + 2\beta\cos\theta_2 & \delta + \beta\cos\theta_2 \\ \delta + \beta\cos\theta_2 & \delta \end{bmatrix}$$

$$= \begin{bmatrix} 20.02 + 12\cos\theta_2 & 4.01 + 6\cos\theta_2 \\ 4.01 + 6\cos\theta_2 & 4.01 \end{bmatrix}$$

② 计算克里斯托费符号 Γ_{ijk}

$$\Gamma_{ijk} = \frac{1}{2}\left(\frac{\partial M_{ij}}{\partial \theta_k} + \frac{\partial M_{ik}}{\partial \theta_j} - \frac{\partial M_{kj}}{\partial \theta_i}\right)$$

$$\frac{\partial M_{11}}{\partial \theta_1} = \frac{\partial M_{12}}{\partial \theta_1} = \frac{\partial M_{22}}{\partial \theta_1} = 0$$

$$\frac{\partial M_{11}}{\partial \theta_2} = -12\sin\theta_2 \qquad \frac{\partial M_{12}}{\partial \theta_2} = -6\sin\theta_2 \qquad \frac{\partial M_{22}}{\partial \theta_2} = 0$$

$$\Gamma_{111} = \frac{1}{2}\left(\frac{\partial M_{11}}{\partial \theta_1} + \frac{\partial M_{11}}{\partial \theta_1} - \frac{\partial M_{11}}{\partial \theta_1}\right) = 0$$

$$\Gamma_{112} = \frac{1}{2}\left(\frac{\partial M_{11}}{\partial \theta_2} + \frac{\partial M_{12}}{\partial \theta_1} - \frac{\partial M_{21}}{\partial \theta_1}\right) = -6\sin\theta_2$$

$$\Gamma_{121} = \frac{1}{2}\left(\frac{\partial M_{12}}{\partial \theta_1} + \frac{\partial M_{11}}{\partial \theta_2} - \frac{\partial M_{21}}{\partial \theta_1}\right) = -6\sin\theta_2$$

$$\Gamma_{122} = \frac{1}{2}\left(\frac{\partial M_{12}}{\partial \theta_2} + \frac{\partial M_{12}}{\partial \theta_2} - \frac{\partial M_{22}}{\partial \theta_1}\right) = -6\sin\theta_2$$

$$\Gamma_{211} = \frac{1}{2}\left(\frac{\partial M_{21}}{\partial \theta_1} + \frac{\partial M_{21}}{\partial \theta_1} - \frac{\partial M_{11}}{\partial \theta_2}\right) = 6\sin\theta_2$$

$$\Gamma_{212} = \frac{1}{2}\left(\frac{\partial M_{21}}{\partial \theta_2} + \frac{\partial M_{22}}{\partial \theta_1} - \frac{\partial M_{12}}{\partial \theta_2}\right) = 0$$

$$\Gamma_{221} = \frac{1}{2}\left(\frac{\partial M_{22}}{\partial \theta_1} + \frac{\partial M_{21}}{\partial \theta_2} - \frac{\partial M_{12}}{\partial \theta_2}\right) = 0$$

$$\Gamma_{222} = \frac{1}{2}\left(\frac{\partial M_{22}}{\partial \theta_2} + \frac{\partial M_{22}}{\partial \theta_2} - \frac{\partial M_{22}}{\partial \theta_2}\right) = 0$$

③ 组装科里奥利与离心力矩阵 $C(\theta, \dot{\theta})$

$$C_{ij} = \sum_{k=1}^{2} \Gamma_{ijk} \dot{\theta}_k$$

$$C_{11} = \Gamma_{111}\dot{\theta}_1 + \Gamma_{112}\dot{\theta}_2 = -6\sin\theta_2 \cdot \dot{\theta}_2$$

$$C_{12} = \Gamma_{121}\dot{\theta}_1 + \Gamma_{122}\dot{\theta}_2 = -6\sin\theta_2(\dot{\theta}_1 + \dot{\theta}_2)$$

$$C_{21} = \Gamma_{211}\dot{\theta}_1 + \Gamma_{212}\dot{\theta}_2 = 6\sin\theta_2 \cdot \dot{\theta}_1$$

$$C_{22} = \Gamma_{221}\dot{\theta}_1 + \Gamma_{222}\dot{\theta}_2 = 0$$

$$C(\theta, \dot{\theta}) = \begin{bmatrix} -6\sin\theta_2 \cdot \dot{\theta}_2 & -6\sin\theta_2(\dot{\theta}_1 + \dot{\theta}_2) \\ 6\sin\theta_2 \cdot \dot{\theta}_1 & 0 \end{bmatrix}$$

④ 计算重力矩阵 $N(\theta)$

$$N_i = \frac{\partial V}{\partial \theta_i}$$

(a) 总势能 $V = 18g\sin\theta_1 + 6g\sin(\theta_1 + \theta_2)$

$$N_1 = \frac{\partial V}{\partial \theta_1} = 18g\cos\theta_1 + 6g\cos(\theta_1 + \theta_2)$$

$$N_2 = \frac{\partial V}{\partial \theta_2} = 6g\cos(\theta_1 + \theta_2)$$

$$N(\theta) = \begin{bmatrix} 18g\cos\theta_1 + 6g\cos(\theta_1 + \theta_2) \\ 6g\cos(\theta_1 + \theta_2) \end{bmatrix}$$

⑤ 得到最终的标准动力学矩阵形式

$$\begin{bmatrix} 20.02 + 12\cos\theta_2 & 4.01 + 6\cos\theta_2 \\ 4.01 + 6\cos\theta_2 & 4.01 \end{bmatrix} \begin{bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} + \begin{bmatrix} -6\sin\theta_2 \cdot \dot{\theta}_1 & -6\sin\theta_2(\dot{\theta}_1 + \dot{\theta}_2) \\ 6\sin\theta_2 \cdot \dot{\theta}_1 & 0 \end{bmatrix}$$

$$+ \begin{bmatrix} 18g\cos\theta_1 + 6g\cos(\theta_1 + \theta_2) \\ 6g\cos(\theta_1 + \theta_2) \end{bmatrix} = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix}$$